Full Molecular Electrolyte Conductivity

This document specifies one computational system that maps an electrolyte recipe, temperature, pressure, and molecular species data to one bulk scalar conductivity. The model resolves the complete periodic molecular liquid. Ion pairs, aggregates, solvent shells, cages, and concentration fluctuations are sampled configurations. They are not configured populations or conductivity corrections.

1 Complete input and output

The physical input is

$$\mathcal{I} = (\mathcal{R}, T, p, \mathcal{S}, \Theta).$$

The recipe $\mathcal{R}$ gives solvent volume ratios, salt molarities, and additive mass fractions. The species data $\mathcal{S}$ give molecular topology, isotope masses, formal molecular charge, interaction sites, and parameters for the common interaction equations. The shared parameter set Θ is fixed over the supported chemical domain.

The numerical arguments are finite-cell scale m, equilibrium integration resolution r, memory resolution ℓ, and basis level n. The complete calculation is

$$\boxed{\begin{aligned}(\mathcal{R}, T, p, \mathcal{S}, \Theta) &\longmapsto \{N_s^{(m)}\}, V_m^\star, q\\ &\longmapsto U_\Theta, \rho_\Theta\\ &\longmapsto K_{\Theta,0}(q), D_\Theta(q), L_\Theta, P\\ &\longmapsto A_n, h_n, C^{\mathrm{dir}}\\ &\longmapsto \sigma_{m,r,\ell,n} \longmapsto \sigma(\mathcal{R}, T, p).\end{aligned}}$$

2 Neutral periodic molecular system

For solvent s, let φ_s be its declared volume ratio, $\bar{v}_s$ its pure-component molar volume, and M_s its molar mass. Choose a reference solvent volume V_0 and set

$$n_s^{\mathrm{solv}} = \frac{\varphi_s V_0}{\bar{v}_s}.$$

For salt formula a at molarity c_a, expand its stoichiometric vector ν_{as} over molecular ions and set

$$n_s^{\mathrm{salt}} = 10^3 V_0 \sum_a \nu_{as} c_a.$$

For additive b with electrolyte mass fraction w_b, solve the exact mass fraction equation

$$n_b M_b = w_b \left[\sum_s n_s^{\mathrm{solv}} M_s + \sum_s n_s^{\mathrm{salt}} M_s + \sum_{b'} n_{b'} M_{b'}\right].$$

This linear system is solved simultaneously for all additives. The resulting molecular amounts define $x_s(\mathcal{R})$. Salts are represented by their component ions; no undissociated population is prescribed.

At cell scale m, choose nonnegative integers

$$\{N_s^{(m)}\} = \arg\min_{\{n_s\}\subset\mathbb{N}_0} \sum_s \frac{1}{\max(x_s,\, m^{-1})} \left(\frac{n_s}{m} - x_s(\mathcal{R})\right)^2$$

subject to

$$\sum_s n_s Q_s = 0, \qquad \mathbf{B}n = \mathbf{b}_m.$$

The first constraint imposes exact molecular electroneutrality. The matrix $\mathbf{B}$ contains exact stoichiometric constraints. The sequence obeys

$$N_s^{(m)}/m \longrightarrow x_s(\mathcal{R}).$$

The integer problem is solved exactly by branch-and-bound. A lexicographic secondary objective on the ordered count vector makes the result deterministic.

Each of the $N_s^{(m)}$ copies instantiates the topology, masses, interaction sites, formal molecular charge, flexible coordinates, and constraint list stored for species s. Indistinguishable copies share parameters and are integrated with the $1/N_s!$ Gibbs factor. Initial molecular conformers are placed in a cubic trial cell by deterministic low-discrepancy translations and rotations. A placement is accepted only when every nonbonded intersite distance exceeds the sum of the corresponding hard initialization radii. Increasing the trial cell is an initialization operation only; the final volume is determined by the NPT measure below. The complete constraint set is denoted $\mathcal{C}$, and the constructed initial state is

$$(\mathcal{R}, T, p, \mathcal{S}, m) \longmapsto (\{N_s^{(m)}\}, H_m^{(0)}, q_m^{(0)}, \mathcal{C}_m).$$

For periodic cell matrix H, volume $V = \det H$, and constrained molecular configuration space Ω_V, define

$$Z_V(T, \{N_s\}) = \int_{\Omega_V} e^{-\beta U_\Theta(q;H)}\, \mathrm{d}q, \qquad \beta = (k_B T)^{-1}.$$

At pressure p,

$$\Delta(T, p, \{N_s\}) = \int_0^\infty e^{-\beta p V} Z_V(T, \{N_s\})\, \mathrm{d}V,$$

$$V_m^\star = \Delta^{-1} \int_0^\infty V e^{-\beta p V} Z_V(T, \{N_s^{(m)}\})\, \mathrm{d}V.$$

Density is therefore an output:

$$\rho_{\mathrm{mass}} = \frac{\sum_s N_s^{(m)} M_s}{N_A V_m^\star}.$$

3 Molecular energy operator

One interaction form applies to every recipe:

$$\boxed{\begin{aligned}U_\Theta = U_{\mathrm{bond}} + U_{\mathrm{angle}} + U_{\mathrm{torsion}} + U_{\mathrm{improper}}\\ + U_{\mathrm{rep}} + U_{\mathrm{disp}} + U_{\mathrm{elec}} + U_{\mathrm{pol}} + U_{\mathrm{mb}}.\end{aligned}}$$

The bonded terms are

$$U_{\mathrm{bond}} = \sum_b \frac{k_b}{2}(r_b - r_b^0)^2, \qquad U_{\mathrm{angle}} = \sum_a \frac{k_a}{2}(\theta_a - \theta_a^0)^2,$$

$$U_{\mathrm{torsion}} = \sum_d \sum_k k_{dk}\left[1 + \cos(n_{dk}\varphi_d - \delta_{dk})\right],$$

$$U_{\mathrm{improper}} = \sum_u \sum_k k_{uk}\left[1 + \cos(n_{uk}\omega_u - \delta_{uk})\right].$$

The topology assigns every bonded tuple exactly once. Holonomic constraints replace, rather than duplicate, the corresponding bonded energy. For the nonbonded realization,

$$U_{\mathrm{rep}} + U_{\mathrm{disp}} = \sum_{i<j} s_{ij}^{\mathrm{LJ}} 4\epsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^6\right],$$

$$\sigma_{ij} = \frac{\sigma_i + \sigma_j}{2}, \qquad \epsilon_{ij} = \sqrt{\epsilon_i \epsilon_j}.$$

The factors are $s_{ij}^{\mathrm{LJ}} = 0$ for 1–2 and 1–3 pairs, $s_{ij}^{\mathrm{LJ}} = s_{14}^{\mathrm{LJ}}$ for 1–4 pairs, and one otherwise. The repulsive r^{-12} term is evaluated in real space. The r^{-6} dispersion term uses the same periodic Ewald decomposition at every cell size:

$$\frac{1}{r^6} = g_\alpha^{(6)}(r) + \frac{1}{V}\sum_{k\neq 0} \widehat{g}_\alpha^{(6)}(k) e^{ik\cdot r} + g_{\mathrm{self}}^{(6)}.$$

The analytical functions $g_\alpha^{(6)}$ and $\widehat{g}_\alpha^{(6)}$ are the real and Fourier transforms obtained by splitting r^{-6} with $e^{-\alpha^2 r^2}$. The same α, real cutoff, reciprocal cutoff, self term, and 1–4 reciprocal correction are used in energy, force, virial, and parameter fitting.

All periodic electrostatic blocks use the same zero-average Green function:

$$-\nabla^2 G_H(x) = 4\pi\left[\sum_{z\in\mathbb{Z}^3} \delta(x - Hz) - V^{-1}\right].$$

Permanent electrostatics are

$$U_{\mathrm{elec}} = \frac{1}{8\pi\varepsilon_0}\sum_{i\neq j} s_{ij}^{\mathrm{C}} q_i q_j G_H(r_i - r_j) + U_{\mathrm{self}}.$$

Here $s_{ij}^{\mathrm{C}} = 0$ for 1–2 and 1–3 pairs, $s_{ij}^{\mathrm{C}} = s_{14}^{\mathrm{C}}$ for 1–4 pairs, and one otherwise. The periodic Green function is evaluated with conducting boundary conditions:

$$\begin{aligned}G_H(r) = &\sum_{n\in\mathbb{Z}^3} \frac{\mathrm{erfc}(\alpha|r + Hn|)}{|r + Hn|}\\ &+ \frac{4\pi}{V}\sum_{k\neq 0} \frac{e^{-k^2/(4\alpha^2)}}{k^2}\cos(k\cdot r) - \frac{\pi}{\alpha^2 V}.\end{aligned}$$

The charge self energy is

$$U_{\mathrm{self}} = -\frac{\alpha}{4\pi^{3/2}\varepsilon_0}\sum_i q_i^2.$$

Scaled exclusions are implemented by adding $(s_{ij}^{\mathrm{C}} - 1) q_i q_j G_H(r_{ij})/(4\pi\varepsilon_0)$, including real, reciprocal, and self-consistent image contributions. The cell is exactly neutral, so no uniform compensating background is present.

For induced dipoles μ, define the periodic field $E_H(q)$ and interaction matrix $T_H(q)$ from derivatives of G_H:

$$U_{\mathrm{pol}}(q, \mu) = \frac{1}{2}\mu^\top \alpha^{-1}\mu - \mu^\top E_H(q) - \frac{1}{2}\mu^\top T_H(q)\mu.$$

For sites i, j, let

$$u_{ij} = \frac{r_{ij}}{(\alpha_i \alpha_j)^{1/6}}, \qquad f_{3,ij} = 1 - e^{-a_{ij} u_{ij}^3}, \qquad f_{5,ij} = 1 - (1 + a_{ij} u_{ij}^3) e^{-a_{ij} u_{ij}^3}.$$

The charge–dipole tensor multiplies the periodic first derivative of G_H by f_3, and the dipole–dipole tensor multiplies its radial and transverse second-derivative components by f_5 and f_3, respectively. The same $a_{ij} = \sqrt{a_i a_j}$ is used in energy and derivatives. Sites with $\alpha_i = 0$ carry no induced dipole. The physical induced dipoles and energy are

$$\mu^\star(q) = \arg\min_\mu U_{\mathrm{pol}}(q, \mu), \qquad U_{\mathrm{pol}}(q) = U_{\mathrm{pol}}(q, \mu^\star(q)).$$

The production model contains no additional U_{mb} closure:

$$\boxed{U_{\mathrm{mb}} \equiv 0.}$$

Many-body induction is already present through the global induced-dipole solve. Any future nonzero many-body interaction defines a different Θ and must replace this equation rather than being appended as an unspecified correction.

All nonbonded parameters are site parameters. Bonded parameters and 1–4 scales come from the common SMIRNOFF assignment over molecular graphs. Permanent charges come from one common electronic-structure charge procedure, and (α_i, a_i) come from the common molecular response fit specified below. Forces and virial derive from this single energy:

$$F_i = -\nabla_{r_i} U_\Theta, \qquad \Xi = -\frac{\partial U_\Theta}{\partial H} H^\top.$$

4 Equilibrium measure

At $V_m^\star$,

$$\boxed{\rho_\Theta(q)\,\mathrm{d}q = Z^{-1} e^{-\beta U_\Theta(q;V_m^\star)}\,\mathrm{d}q.}$$

Rigid constraints $g(q) = 0$ restrict this measure to the constraint manifold with its induced volume element. Equilibrium integration uses a sequence $\mathcal{Q}_r$ with

$$\mathcal{Q}_r[f] \longrightarrow \langle f \rangle_{\rho_\Theta} = \int f(q)\rho_\Theta(q)\,\mathrm{d}q.$$

Pair basins, coordination shells, aggregates, cages, and ligand states are smooth observables of q. They do not alter ρ_Θ.

4.1 NPT integration

Use fractional molecular coordinates s, an upper-triangular cell matrix H with positive diagonal, constrained momenta p, cell momentum P_H, and barostat mass W_H. The sampled extended density is

$$\pi_{\mathrm{NPT}}(s, p, H, P_H) \propto (\det H)^{N_m} \delta(g)\delta(\dot g) \times \exp\left\{-\beta\left[\mathcal{H}_\Theta(Hs, p) + p_{\mathrm{ext}} \det H + \frac{\mathrm{tr}(P_H^\top P_H)}{2W_H}\right]\right\}.$$

The factor $(\det H)^{N_m}$ is the molecular-coordinate volume Jacobian. Translations, rotations, and internal flexible moves use Metropolis proposals in fractional coordinates. The cell proposal is

$$H' = H\exp(\varepsilon S),$$

where S is a symmetric trace-plus-shear proposal. The acceptance probability is

$$\min\left\{1, \exp[-\beta(\Delta U + p_{\mathrm{ext}}\Delta V)]\left(\frac{V'}{V}\right)^{N_m} \frac{J(H' \to H)}{J(H \to H')}\right\}.$$

Constraint moves use RATTLE projection and the induced constraint measure. A Metropolis correction removes finite-step bias. Independent chains start from distinct low-discrepancy placements and Maxwell-distributed tangent momenta.

For integration level r, halve every proposal scale and double accepted samples, equilibration sweeps, and independent chains:

$$\delta_r = \delta_0 2^{-r}, \quad N_r = N_0 2^r, \quad R_r = R_0 + r.$$

The stationary suffix of energy, volume, density, and each retained slow coordination observable defines the production average. The equilibrium volume is the mean stationary volume. Configurations are mapped to that mean cell in fractional coordinates before fixed-volume Dirichlet quadrature. This sequence converges to the NPT measure because every proposal is reversible, irreducible on each connected constraint component, and Metropolis corrected.

5 Molecular coordinates and null modes

For molecule a,

$$R_a = \frac{1}{M_a}\sum_{i\in a} m_i r_i, \quad V_a = \frac{1}{M_a}\sum_{i\in a} m_i v_i, \quad F_a = \sum_{i\in a} F_i.$$

Collect the molecular translational and rotational velocities in V, with mass–inertia matrix M. Let the columns of $B(q)$ span uniform translation and exact rigid-constraint null modes. The physical-range projector in the kinetic-energy metric is

$$\boxed{\Pi(q) = I - B(q)\left[B(q)^\top M B(q)\right]^\# B(q)^\top M.}$$

Memory, friction, diffusion, and Poisson operators act on $\operatorname{ran}\Pi$.

6 Microscopic dynamics for molecular memory

The memory kernel is generated before the configuration diffusion and therefore cannot be evolved by L_Θ. Introduce constrained atomic momenta p, mass matrix M_{at}, and the phase-space Hamiltonian

$$\mathcal{H}_\Theta(q, p) = \frac{1}{2} p^\top M_{\mathrm{at}}^{-1} p + U_\Theta(q).$$

Write $G(q) = \nabla_q g(q)$, $v = M_{\mathrm{at}}^{-1} p$, and $A_g(q) = G(q) M_{\mathrm{at}}^{-1} G(q)^\top$. On the cotangent bundle of the molecular constraint manifold,

$$\dot q = v, \qquad \dot p = -\nabla_q U_\Theta - G(q)^\top \lambda,$$

where the acceleration constraint determines the multiplier without an additional physical parameter:

$$\lambda = A_g(q)^\#\left[\dot G(q; v)v - G(q) M_{\mathrm{at}}^{-1} \nabla_q U_\Theta\right].$$

The microscopic generator is therefore

$$\boxed{\mathcal{G}_\Theta f = (M_{\mathrm{at}}^{-1} p)^\top \nabla_q f - \left(\nabla_q U_\Theta + G^\top \lambda\right)^\top \nabla_p f,}$$

on $g(q) = 0$ and $G(q) M_{\mathrm{at}}^{-1} p = 0$. Its invariant canonical measure is

$$\boxed{\varrho_\Theta(q, p)\,\mathrm{d}q\,\mathrm{d}p = \mathcal{Z}^{-1} e^{-\beta\mathcal{H}_\Theta(q,p)}\,\mathrm{d}\lambda_g(q)\,\mathrm{d}\lambda_{T_q}(p).}$$

Here λ_g and λ_{T_q} are the induced configuration and tangent momentum measures. This phase-space dynamics uses exactly the energy, masses, cell, and constraints that define ρ_Θ. It is used to evaluate molecular memory; it is not the later homogenized configuration process.

7 Conditional memory and diffusion

Let

$$\xi = \Xi(q)$$

contain the resolved molecular centers, orientations, and the complete hierarchy of local density, coordination, angular, and charge-density observables defined below. The symbol q continues to denote the full constrained atomic configuration. Let $\mathcal{P}_\Xi$ be conditional expectation in $L^2(\varrho_\Theta)$ onto (ξ, V), where V contains the resolved conjugate translational and rotational velocities. Set $\mathcal{Q}_\Xi = I - \mathcal{P}_\Xi$. The orthogonal molecular force and its orthogonal evolution are

$$F^\perp = \mathcal{Q}_\Xi F, \qquad F^\perp(t) = e^{t\mathcal{Q}_\Xi \mathcal{G}_\Theta \mathcal{Q}_\Xi} F^\perp.$$

The fluctuation–dissipation definition of the conditional memory kernel is

$$\boxed{K_\Theta(\xi, t) = \beta\,\mathbb{E}_{\varrho_\Theta}\left[\Pi F^\perp(t) F^\perp(0)^\top \Pi \mid \Xi(q(0)) = \xi\right].}$$

The conditional expectation integrates canonical momenta and all atomic coordinates within the fiber $\Xi^{-1}(\xi)$. Thus K_Θ is determined by $(U_\Theta, M_{\mathrm{at}}, \varrho_\Theta, \mathcal{P}_\Xi)$, independently of D_Θ and L_Θ. Define

$$C_{VV}(\xi, t) = \mathbb{E}[\Pi V(t) V(0)^\top \Pi \mid \Xi(q(0)) = \xi], \qquad C_{PV}(\xi, t) = M C_{VV}(\xi, t).$$

These obey the matrix Volterra equation

$$\frac{\partial C_{PV}(\xi, t)}{\partial t} = -\int_0^t K_\Theta(\xi, \tau) C_{VV}(\xi, t - \tau)\,\mathrm{d}\tau.$$

For $s > 0$,

$$\boxed{\widetilde K_\Theta(\xi, s) = \left[C_{PV}(\xi, 0) - s\widetilde C_{PV}(\xi, s)\right]\widetilde C_{VV}(\xi, s)^\#.}$$

The zero-frequency friction and molecular diffusion are

$$\boxed{\begin{aligned} K_{\Theta,0}(\xi) &= \lim_{s\downarrow 0} \Pi \widetilde K_\Theta(\xi, s)\Pi, \\ D_\Theta(q) &= k_B T\,\Pi K_{\Theta,0}(\Xi(q))^\# \Pi. \end{aligned}}$$

The finite production representation is a positive matrix exponential expansion. For N_m molecules, order generalized molecular velocities as

$$\mathsf{V} = (V_1, \Omega_1, \ldots, V_{N_m}, \Omega_{N_m}) \in \mathbb{R}^{6N_m}.$$

For a linear molecule, remove rotation about its molecular axis. For a point ion, remove all rotational coordinates. Let d_m be the resulting generalized dimension. The kernel is

$$\boxed{K_\Theta(q, t) = \Pi(q)\sum_{\ell=0}^{L} B_{\Theta,\ell}(q) B_{\Theta,\ell}(q)^\top e^{-t/\tau_\ell}\Pi(q),}$$

$$\tau_\ell = \tau_{\min}\left(\frac{\tau_{\max}}{\tau_{\min}}\right)^{\ell/L}, \qquad \ell = 0, \ldots, L.$$

The endpoints $\tau_{\min}$ and $\tau_{\max}$ are the trajectory sampling interval and retained kernel horizon. Increasing ℓ refines this fixed interval; it does not change the physical model.

The columns of $B_{\Theta,\ell}$ are generated from one geometry map. For molecules a, b, define

$$r_{ab} = |R_b - R_a|_H, \qquad \hat r_{ab} = (R_b - R_a)_H / r_{ab},$$

$$R_k(r) = \begin{cases} \sqrt{2/r_c}\,\sin(k\pi r/r_c)\,f_c(r), & 0 < r < r_c, \\ 0, & r \ge r_c, \end{cases} \qquad f_c(r) = \frac{1}{2}[1 + \cos(\pi r/r_c)].$$

The molecular environment descriptor contains

$$\begin{aligned} \zeta_a(q) = \{&M_a,\ Q_a,\ \mathcal{I}_a,\ \alpha_a,\ \sum_{b\ne a} R_k(r_{ab})\,s_b, \\ &\sum_{b\ne a} R_k(r_{ab}) Y_{\lambda m}(\hat r_{ab})\,s_b,\ \sum_{b,c\ne a} R_k(r_{ab}) R_{k'}(r_{ac}) P_\lambda(\hat r_{ab}\cdot\hat r_{ac}) s_b s_c, \\ &\rho_a^{(k)},\ \rho_{Q,a}^{(k)},\ \mathcal{O}_a^{(\lambda)},\ \mathcal{G}_a^{(d)}\}. \end{aligned}$$

Here $s_b = (1, Q_b/e, M_b/M_0, \alpha_b/\alpha_0)$; $\rho_a^{(k)}$ and $\rho_{Q,a}^{(k)}$ are local number and charge densities; $\mathcal{O}_a^{(\lambda)}$ are molecular orientation invariants; and $\mathcal{G}_a^{(d)}$ are powers through depth d of the smooth adjacency

$$\mathcal{A}_{ab}^{(k)} = R_k(r_{ab})\left[1 + \exp\{-\gamma(C_{ab} - C_0)\}\right]^{-1},$$

with smooth coordination

$$C_{ab} = \sum_{i\in a}\sum_{j\in b}\left[1 + (r_{ij}/r_0)^p\right]^{-1}.$$

All constants in these expressions belong to Θ_{memory}. Atomic numbers, masses, charges, polarizabilities, topology, and geometry enter the map. Species names and recipe indices do not.

At hierarchy level n, enumerate all translationally invariant, rotation-equivariant matrix features $\{\mathsf{E}_\nu^{(n)}(q)\}_{\nu=1}^{p_n}$ generated from the listed descriptors with radial order, angular order, graph depth, and product degree at most n. The enumeration is cumulative, so $\mathsf{E}^{(n)} \subset \mathsf{E}^{(n+1)}$. For generalized molecular velocity dimension d_m, define the memory factors explicitly by

$$B_{\Theta,\ell}^{(n)}(q) = \sum_{\nu=1}^{p_n} \theta_{\ell\nu} \mathsf{E}_\nu^{(n)}(q), \qquad B_{\Theta,\ell}^{(n)}(q) \in \mathbb{R}^{d_m \times d_m}.$$

The coefficient tensors $\theta_{\ell\nu}$ are the learned entries of Θ_{memory}. Self-block features depend on ζ_a; cross-block features depend on $(\zeta_a, \zeta_b, R_k(r_{ab}), Y_{\lambda m}(\hat{r}_{ab}))$ and satisfy

$$B_{ab,\ell}(q) = B_{ba,\ell}(q)^\top.$$

Permutation-equivariant assembly of these blocks produces $B_{\Theta,\ell}^{(n)}(q)$. Multiplication by its transpose gives a positive semidefinite kernel before null-space projection. The projector removes uniform translation, constrained rotations, and every holonomic constraint velocity:

$$\Pi(q) = I - \mathsf{B}(q)\left[\mathsf{B}(q)^\top \mathsf{M}\mathsf{B}(q)\right]^\# \mathsf{B}(q)^\top \mathsf{M}.$$

The zero-frequency operators are therefore constructive:

$$\boxed{\begin{aligned} K_{\Theta,0}(q) &= \Pi(q) \sum_{\ell=0}^{L} \tau_\ell B_{\Theta,\ell}(q) B_{\Theta,\ell}(q)^\top \Pi(q), \\ D_\Theta(q) &= k_B T\, \Pi(q) K_{\Theta,0}(q)^\# \Pi(q). \end{aligned}}$$

For each force-complete trajectory, evaluate $\xi_n = \Xi(q_n)$, fit the conditional mean force on the training frames only, and form

$$\widehat{F}_n^\perp = \Pi_n \left[F_n - \widehat{\mathbb{E}}(F \mid \xi_n, \mathsf{V}_n)\right].$$

Let $w_b(\xi)$ be a normalized compactly supported partition of unity on descriptor space, refined by level b. The finite conditional target is

$$\boxed{\widehat{K}_b(\xi, t_j) = \beta \frac{\sum_n w_b(\xi - \xi_n) \widehat{F}_{n+j}^\perp \widehat{F}_n^{\perp\top}}{\sum_n w_b(\xi - \xi_n)}.}$$

Lag pairs crossing trajectory or replica boundaries are excluded. Self and cross blocks are retained in one matrix. Conditional cells below the configured effective sample count are unresolved and trigger additional sampling. They are never replaced by a recipe average. With recipe-disjoint training set $\mathcal{T}$ and held-out set $\mathcal{V}$, fit

$$\Theta^\star_{\text{memory}} = \arg\min_\Theta \sum_{r \in \mathcal{T}} \sum_{b,j} \frac{N_{rbj}}{\|\widehat{K}_b^{(r)}(0)\|_F^2} \left\| W_{rb}^{1/2} \left[K_\Theta(\xi_{rb}, t_j) - \widehat{K}_b^{(r)}(t_j)\right] W_{rb}^{1/2} \right\|_F^2.$$

W_{rb} is the inverse covariance of the molecular generalized velocities after removing null modes. Rank-revealing SVD removes descriptor columns and target directions below the numerical singular-value tolerance. Model selection uses only held-out kernel error, integrated-friction error, and force-correlation error. Measured conductivity never enters this objective.

The independent zero-frequency mobility check uses the positive covariance-growth form

$$\boxed{D_{\Theta,\ell}(\xi) = \frac{1}{2t_\ell} \mathbb{E}\left[\Delta R_\ell \Delta R_\ell^\top \mid \Xi(q(0)) = \xi\right], \qquad K_{\Theta,0,\ell} = k_B T\, D_{\Theta,\ell}^\#,}$$

where $\Delta R_\ell = \int_0^{t_\ell} \Pi V(t)\, dt$ and the conditional mean displacement is removed. This estimator validates the zero-frequency integral. Lag-resolved orthogonal-force targets remain the source of the fitted kernel. This form is positive semidefinite at every finite resolution and converges to the same zero-frequency mobility as the Laplace–Volterra expression when the velocity correlation is integrable. It avoids replacing a noisy indefinite kernel estimate by clipped eigenvalues. The limit is evaluated on the physical range. Conserved and explicitly resolved slow coordinates remain outside the inversion.

The replacement of the non-Markovian molecular equation by $D_\Theta(q)$ is its zero-frequency homogenization. When the orthogonal-force kernel is integrable and the eliminated variables mix on a finite time scale, this homogenized operator has the same zero-frequency molecular mobility as the generalized Langevin equation.

8 Constrained reversible generator

Let T_q project Cartesian molecular increments onto the tangent space of the rigid constraints. Replace D_Θ by $T_q D_\Theta T_q^\top$ and retain the same symbol. The production generator is

$$\boxed{L_\Theta f = \rho_\Theta^{-1} \nabla_q \cdot (\rho_\Theta D_\Theta \nabla_q f).}$$

In coordinates,

$$L_\Theta f = -\beta \nabla U_\Theta^\top D_\Theta \nabla f + (\nabla \cdot D_\Theta)^\top \nabla f + D_\Theta : \nabla^2 f.$$

The associated stochastic dynamics is

$$dq_t = [-\beta D_\Theta \nabla U_\Theta + \nabla \cdot D_\Theta]\, dt + \sqrt{2D_\Theta}\, dW_t.$$

The divergence drift and correlated noise belong to the same operator. For smooth constraint-compatible functions,

$$-\langle f, L_\Theta g \rangle_{\rho_\Theta} = \left\langle \nabla f^\top D_\Theta \nabla g \right\rangle_{\rho_\Theta}.$$

Thus L_Θ is reversible and negative semidefinite on mean-zero functions.

9 Charge polarization

Use formal molecular charges and unwrapped molecular centers:

$$\boxed{P(q) = \sum_a Q_a R_a^{\text{unwrap}}.}$$

Molecular translation carries unbounded charge displacement. Intramolecular rotation, vibration, and induced polarization are bounded unless molecular charge topology changes. The drift current is

$$j_i(q) = L_\Theta P_i(q).$$

The stochastic quadratic variation supplies

$$\boxed{C_{ij}^{\text{dir}} = \left\langle \nabla P_i^\top D_\Theta \nabla P_j \right\rangle_{\rho_\Theta}.}$$

10 Poisson corrector and scalar conductivity

For each Cartesian component solve

$$\boxed{-L_\Theta \chi_i = L_\Theta P_i, \qquad \langle \chi_i \rangle_{\rho_\Theta} = 0.}$$

The weak equation is

$$\left\langle \nabla \varphi^\top D_\Theta \nabla \chi_i \right\rangle = -\left\langle \nabla \varphi^\top D_\Theta \nabla P_i \right\rangle$$

for every admissible mean-zero test function φ. The homogenized charge diffusion is

$$\boxed{\mathcal{D}_{ij}^Q = C_{ij}^{\text{dir}} - \left\langle L_\Theta P_i, (-L_\Theta)^\# L_\Theta P_j \right\rangle_{\rho_\Theta}.}$$

Equivalently,

$$\mathcal{D}_{ii}^Q = \min_{\chi \in \mathcal{H}_E} \left\langle (\nabla P_i + \nabla \chi)^\top D_\Theta (\nabla P_i + \nabla \chi) \right\rangle_{\rho_\Theta}.$$

With P measured in C m,

$$\boxed{\sigma(\mathcal{R}, T, p) = \frac{1}{3k_B T V_m^\star} \operatorname{tr} \mathcal{D}^Q \quad [\mathrm{S\,m^{-1}}].}$$

11 Constructive nested basis

At level n, use cutoff $r_c^{(n)} = \min(r_{\max}, n r_0)$, radial order $1 \le k \le n$, angular order $0 \le \lambda \le n$, reciprocal vectors

$$\mathcal{K}_n = \{2\pi H^{-\top} z : z \in \mathbb{Z}^3,\ 0 < |z|_\infty \le n\},$$

graph depth $0 \le d \le n$, and product degree $1 \le d_p \le n$. The primitive scalar functions are:

$$\rho_{a,k,s} = \sum_{b \ne a} R_k(r_{ab}) s_b, \qquad c_{a,k\lambda} = \sum_{b,c \ne a} R_k(r_{ab}) R_k(r_{ac}) P_\lambda(\hat{r}_{ab} \cdot \hat{r}_{ac}),$$

$$\rho_z = \sum_a e^{iz \cdot R_a}, \qquad \rho_z^Q = \sum_a Q_a e^{iz \cdot R_a}, \qquad z \in \mathcal{K}_n,$$

$$o_{a,\lambda\lambda'} = \sum_{m=-\lambda}^{\lambda} \mathcal{O}_{a,\lambda m} \mathcal{O}_{a,\lambda' m}^*, \qquad g_{a,k,d} = (\mathcal{A}_k^d \mathbf{1})_a.$$

Coordination functions use

$$c_{ab}^{(k)} = \sum_{i \in a, j \in b} R_k(r_{ij}),$$

and include powers and cross-products through degree n. Real and imaginary Fourier parts are stored separately after removing the zero wavevector. Molecular orientation tensors $\mathcal{O}_{a,\lambda m}$ are spherical harmonics of the principal inertia frame, symmetrized over degenerate axes. Point ions have only the $\lambda = 0$ component.

Order primitive functions lexicographically by family, cutoff, radial order, angular order, wavevector, graph depth, and molecular index. Form every symmetry-compatible monomial of total degree at most n. Sum molecular-indexed monomials over indistinguishable molecules. Center each function with its equilibrium mean and remove columns whose Dirichlet norm vanishes or whose rank-revealing QR pivot is below the level tolerance. Denote the surviving ordered set by $\mathcal{G}_n$. Since level $n+1$ repeats every index range from level n before adding new functions,

$$V_n = \operatorname{span} \mathcal{G}_n, \qquad V_n \subset V_{n+1}.$$

Smooth compactly supported radial functions, trigonometric polynomials on the periodic cell, spherical harmonics, and their products form a separating algebra on every compact collision-free constraint component. Stone–Weierstrass density, followed by smooth approximation in the weighted Sobolev norm, gives

$$\overline{\bigcup_{n=1}^{\infty} V_n}^{\|\cdot\|_E} = \mathcal{H}_E, \qquad \|f\|_E^2 = \langle \nabla f^\top D_\Theta \nabla f \rangle.$$

For basis $\Phi_n = (\phi_1, \dots, \phi_{d_n})$, compute

$$A_{\mu\nu}^{(n)} = \left\langle \nabla \phi_\mu^\top D_\Theta \nabla \phi_\nu \right\rangle,$$

$$h_{\mu i}^{(n)} = \left\langle \nabla \phi_\mu^\top D_\Theta \nabla P_i \right\rangle.$$

The coefficient vector solves

$$\boxed{A^{(n)} c_i^{(n)} = -h_i^{(n)}.}$$

The finite charge diffusion and conductivity are

$$\boxed{\mathcal{D}_{ij}^{Q,n} = C_{ij}^{\text{dir}} - (h_i^{(n)})^\top (A^{(n)})^\# h_j^{(n)},}$$

$$\boxed{\sigma_{m,r,\ell,n} = \frac{\operatorname{tr}[C^{\text{dir}} - h_n^\top A_n^\# h_n]}{3k_B T V_m^\star}.}$$

12 Residual-driven enrichment

Split independent equilibrium chains into fit and held-out sets before evaluating any basis matrix. On held-out configuration q, evaluate the generator action analytically:

$$L_\Theta\phi = -\beta\nabla U_\Theta^\top D_\Theta\nabla\phi + (\nabla\cdot D_\Theta)^\top\nabla\phi + D_\Theta : \nabla^2\phi.$$

The divergence is obtained by automatic differentiation of the matrix kernel representation, and the Hessian-vector contraction is evaluated without forming the full Hessian. For the fitted corrector $\chi_{n,i} = \sum_\mu c_{\mu i}\phi_\mu$, the held-out residual is

$$r_{n,i}(q) = L_\Theta P_i(q) + L_\Theta\chi_{n,i}(q).$$

Its graph norm is

$$\|r_n\|^2_{-1,n+1} = \sum_{i=1}^{3} b_{n,i}^\top A_{n+1}^\# b_{n,i}, \qquad (b_{n,i})_\mu = \langle\phi_\mu, r_{n,i}\rangle_{\text{heldout}}.$$

For $g \in V_{n+1}\setminus V_n$, define

$$\widetilde{A}_{gg} = A_{gg} - A_{g\Phi}A_{\Phi\Phi}^\# A_{\Phi g},$$
$$\widetilde{h}_{g,i} = h_{g,i} - A_{g\Phi}A_{\Phi\Phi}^\# h_{\Phi,i}.$$

When $\widetilde{A}_{gg} > 0$, the exact gain in the variational correction is

$$S(g) = \sum_{i=1}^{3}\frac{|\widetilde{h}_{g,i}|^2}{\widetilde{A}_{gg}}.$$

Candidates come from the next declared hierarchy level. Exhausting one finite dictionary does not establish convergence. Select the candidate with maximum held-out $S(g)$, refit on the fit chains, and recompute the held-out residual. When all candidates at level $n+1$ fall below the residual tolerance, construct the complete level $n+2$. Stop only when the held-out graph norm and the conductivity change both pass their tolerances on two successive hierarchy levels.

13 Galerkin convergence

Let $\chi_i^\star$ solve the full Poisson problem and $\chi_{n,i}$ be its Galerkin projection. Galerkin orthogonality gives

$$\langle\nabla(\chi_i^\star - \chi_{n,i})^\top D_\Theta\nabla\phi\rangle = 0, \qquad \phi\in V_n.$$

Expanding the variational energy gives

$$\boxed{\sigma_{m,r,\ell,n} - \sigma_{m,r,\ell,\infty} = \frac{1}{3k_BTV_m^\star}\sum_{i=1}^{3}\|\chi_i^\star - \chi_{n,i}\|_E^2 \geq 0.}$$

Therefore

$$\sigma_{m,r,\ell,n}\downarrow\sigma_{m,r,\ell,\infty}.$$

When equilibrium integration converges, the zero-frequency memory limit exists on the physical range, the basis hierarchy is energy-norm dense, and the thermodynamic limit exists,

$$\boxed{\sigma(\mathcal{R},T,p) = \lim_{m\to\infty}\lim_{r\to\infty}\lim_{\ell\to\infty}\lim_{n\to\infty}\sigma_{m,r,\ell,n}.}$$

For each finite cell, the basis limit $n\to\infty$ is taken first, followed by the memory-resolution limit $\ell\to\infty$, the equilibrium-integration limit $r\to\infty$, and finally the composition-preserving thermodynamic limit $m\to\infty$. Every limit refines a named numerical approximation while U_Θ, K_Θ, P, T, p, and the intensive recipe remain fixed.

14 Shared parameter set

The complete parameter vector is

$$\boxed{\Theta = (\Theta_{\text{bonded}}, \Theta_{\text{nonbonded}}, \Theta_{\text{electrostatic}}, \Theta_{\text{polarization}}, \Theta_{\text{memory}}).}$$

Θ_{bonded} contains, in SI units, atom-pattern assignments, (k_b, r_b^0), (k_a, θ_a^0), and every (k,n,δ) tuple for proper and improper torsions. $\Theta_{\text{nonbonded}}$ contains site (σ_i, ϵ_i), the two 1–4 scales, dispersion Ewald splitting, and real and reciprocal cutoffs. $\Theta_{\text{electrostatic}}$ contains permanent site charges, Coulomb Ewald splitting, and its cutoffs. $\Theta_{\text{polarization}}$ contains site polarizability tensors, Thole parameters, and induced-dipole solve tolerance. Θ_{memory} contains descriptor cutoffs, radial and angular orders, coordination switching parameters, graph depth, correlation-product order, decay times, and every coefficient of $B_{\Theta,\ell}$.

Bonded and nonbonded interaction parameters are fit once to ωB97M–V/def2–TZVPPD electronic-structure energies, forces, optimized geometries, torsion scans, electrostatic potentials, finite-field dipole response, molecular polarizabilities, and virials. Counterpoise correction is applied to intermolecular dimers. Permanent charges are MBIS charges constrained to the formal molecular charge. For reference configurations x_r, minimize

$$\mathcal{L}_U = w_E\sum_r|U_\Theta(x_r) - E_r|^2 + w_F\sum_r\|\nabla U_\Theta(x_r) + F_r\|^2$$
$$+ w_\Xi\sum_r\|\Xi_\Theta(x_r) - \Xi_r\|_F^2 + w_\mu\sum_r\|\mu_\Theta(x_r) - \mu_r\|^2.$$

The weights are $w_E = (N_{\text{at}}E_0^2)^{-1}$, $w_F = (3N_{\text{at}}F_0^2)^{-1}$, $w_\Xi = (9\Xi_0^2)^{-1}$, and $w_\mu = (3\mu_0^2)^{-1}$, where each zero-subscript scale is the root-mean-square magnitude of that target over the training split. The electronic-structure chemistry split is by complete molecule, so held-out molecules do not contribute configurations to the fit.

The memory parameters use the force-correlation objective given above. Its split is by complete recipe, concentration, and temperature. A molecule or recipe may appear in only one split. The supported domain is the closure of elemental, topological, charge, temperature, pressure, and descriptor ranges represented in the interaction and memory training sets. Inputs outside any range terminate with an unsupported-domain result. They do not select alternate equations.

Fixed parameters are physical constants, molecular topology, isotope masses, formal molecular charges, the interaction functional forms, periodic boundary conditions, and null-space definitions. Learned parameters are the numerical entries of the five blocks of Θ. Experimental conductivity is reserved for external evaluation of the completed map:

$$(\mathcal{R}, T, p, \mathcal{S}, \Theta)\longmapsto\sigma.$$

It is absent from $\mathcal{L}_U$, the memory loss, basis selection, and every intermediate operator.

15 Executable calculation order

1. Decode $\mathcal{R}$ and construct neutral integer counts.
2. Evaluate U_Θ and equilibrate at (T,p) to obtain $V_m^\star$ and samples of ρ_Θ.
3. Construct $\mathcal{G}_\Theta$ from the same energy, masses, cell, and molecular constraints; sample its canonical phase-space measure.
4. Project atomic coordinates, momenta, velocities, and forces to constrained molecular variables and form $\mathcal{P}$ and $\mathcal{Q}$.
5. Evolve orthogonal forces with $e^{t\mathcal{Q}\mathcal{G}_\Theta\mathcal{Q}}$, evaluate their conditional correlations, and solve for $K_{\Theta,0}(q)$ on the physical range.
6. Form $D_\Theta(q) = k_BTK_{\Theta,0}(q)^\#$.
7. Evaluate C^{dir}, A_n, and h_n over the same equilibrium measure.
8. Solve $A_nc_{n,i} = -h_{n,i}$ after removing exact null modes.
9. Return
$$\boxed{\sigma_{m,r,\ell,n} = \frac{\operatorname{tr}(C^{\text{dir}} - h_n^\top A_n^\# h_n)}{3k_BTV_m^\star}.}$$
10. Refine (m,r,ℓ,n) without changing the physical model.

16 Model boundary

The equations contain no configured SSIP or CIP fractions, mass-action population allocation, species conductivity bonus, terminal conductivity multiplier, or recipe-specific transport branch. Species data determine molecular construction and interaction parameters. The equilibrium measure determines structure. Conditional memory determines diffusion. The Poisson corrector determines correlated charge transport. Their composition returns the scalar conductivity.